MIL-R-11019B
24 MAY 1963
SUPERSEDING
MIL-R-11019A
24 DECEMBER 1952

MILITARY SPECIFICATION

REEL UNIT RL-128( )/G

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of manually operated metal payout reel which is adjustable to fit different sizes of coils of open wire and which has an automatic brake that operates from the application or release of tension on the payed-out wire.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 —Bearings; Ball.
TT-C-595 —Colors; (for) Ready-Mixed Paints.

MILITARY

MIL-P-116 —Preservation, Methods of.
JAN-P-125 —Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN-P-132 —Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed.
MIL-W-12883 — Welding; Joint Design, Data for.
MIL-M-13231 —Marking of Electronic Items.

U. S. ARMY

72-58 —Finishes (for Ground Signal Equipment.)

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.
MIL-STD-130 — Identification Marking of U. S. Military Property.
MIL-STD-169 — Extreme - Temperature Cycle.
MIL-STD-170 — Moisture Resistance Test Cycle for Ground Signal Equipment.

DRAWINGS

SIGNAL CORPS

SC-DL-107577 —Drawing and Data List for Payout Reel, Cable RL-128 ( )/G.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 **Other publications.**—The following documents form a part of this specification. Unless otherwise stated, the issue in effect on date of invitation for bids form a part of this specification:

NATIONAL BUREAU OF STANDARDS
Handbook H-28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

ASSOCIATION OF AMERICAN RAILROADS
Uniform Freight Classification Ratings, Rules, and Container Specifications for Rail Shipments.

(Application for copies of the above publication should be addressed to Association of American Railroads, Chicago, Illinois.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules and Container Specifications for Truck Shipments.

(Application for copies of the above publication should be addressed to American Trucking Association, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Construction. — Construction of the equipment shall be in accordance with the drawings listed on Drawing SC-DL-107577 and shall meet the requirements of this specification. In case of conflict between the specification and drawings, the specification shall govern.

3.2 Preproduction samples.

3.2.1 *Quantities* —After award of contract, two preproduction samples shall be furnished for Government inspection and approval.

3.2.2 *Compliance with specified requirements.*

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production and shall be accompanied by a statement to that effect. (See 3.2.2.3 for exception.)

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.4) preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 *Test data* — The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like.)

3.2.4 *Reference standards.*—After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.3 Parts, materials, and processes.

3.3.1 *Ball bearings* — Ball bearings used

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shall be in accordance with Specification FF-B-171

3.3.2 *Metals.*—Metals used shall be of corrosion resisting types or shall be treated in accordance with Specification 72-53. Dissimilar metals used in contact with each other shall be carefully selected in accordance with Specification 72-53 to prevent electrolytic corrosion in the presence of moisture.

3.3.3 *Welding.*—Welds shall be of ample size Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Welded joints shall conform to Specification MIL-W-12683, and shall be such that grinding on the finished weld will be unnecessary.

3.3.3.1 Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.3.3.2 *Process* —Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved.

3.3.4 *Cleaning.*—Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.3.5 *Screws, nuts, bolts and tapped holes.*—Unless otherwise allowed, all screws, bolts, tapped holes, and nuts shall have the American National form of thread in accordance with the National Bureau of Standards Handbook H28 and shall be Class 2 fit.

3.3.5.1 *Means of locking.*—Nuts and bolts shall be secured by lockwashers, lock nuts, stop nuts, castellated nuts with cotter pins, or safety wires. Lock nuts, as made by the Elastic Stop Nut Company, Union, New Jersey, or equal, shall be used wherever practical. Locking methods shall be capable of withstanding repeated use and shall insure locking under severest conditions of shock and vibration encountered in service conditions as specified in Section 4. Staking may be used in lieu of the foregoing methods, where parts are designed to be permanently fastened.

3.3.6 *Finishes.*—Where not otherwise specified, the equipment shall be given protective finish in accordance with Specification 72-53. Paint finish shall be semi-gloss enamel conforming to Number 2430 of Specification TT-C-595 (green color, equivalent to olive-drab.)

3.3.7 *Ruggedness and reliability.* — All parts and details of construction shall have the maximum of ruggedness and reliability under all operating conditions. The equipment shall be so constructed that it will withstand long periods of service and repeated rough handling in transportation, with a minimum of damage or need of adjustment.

3.4 Operations. — The brake mechanism shall function automatically by the application or release of tension on payed-out wire. The reel shall operate freely without vibration and there shall be no backlash of wire when tested in accordance with 4.6 The

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brake mechanism shall also function by manual operation of the brake lever and shall meet the test of 4.7

3.5 **Temperature and moisture.**—The equipment shall be capable of satisfactory operation after storage or transportation at any temperature within the range from —80°F to + 160°F., and at any relative humidity up to 100 percent Full specification requirements shall be met in any external ambient temperature within the range —40°F. to + 131°F All items or parts entering into the construction of any unit of the complete equipment shall be capable of operating continuously, without damage, at any external ambient temperature within the operating range specified herein.

3.6 **Interchangeability.** — All components, subassemblies and parts shall be mechanically interchangeable with all corresponding components, subassemblies and parts of the equipment or equipments on order

3.7 **Marking.**

3.7.1 *General.*—Marking shall conform to Specification MIL–M–13231 except that the notation "U.S." shall be replaced with "Signal Corps, U.S. Army" for Department of the Army procurement

3.7.1.1 *Air Force* —For Air Force procurement equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL–STD–130 The identification data appearing on the reel nameplate shall be as follows

Reel Unit RL–128( )/G
Stock Number.. ....
Manufacturer's Identification
Inspector's Stamp
U.S

3.7.2 *Serial numbers.*—Serial numbers under Specification MIL–M–13231 are not required

3.8 **Workmanship.**—The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein:

3.3.3 **Welding (including subparagraphs 3.3.3.1 and 3.3.3.2).**
3.3.4 **Cleaning.**
3.3.5.1 **Securing of parts.**
3.3.6 Finishes.
3.7 Marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Classification of inspection.**—Inspection shall be classified as follows·

(a) Preproduction inspection

(b) Acceptance inspection

4.2 **Preproduction inspection.**—Preproduction inspection shall consist of the group A inspection specified in Table II and the preproduction inspection specified in Table I Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection may be performed in any order.

TABLE I. *Preproduction inspection*

Inspection (For additional preproduction inspection see 4 2)	Req. par	Test par
Moisture Resistance..	3.5	4.9
Temperature	3 5	4.8

4.3 **Acceptance inspection.** — Acceptance inspection shall consist of group A inspection as specified in 4.3.1. Each unit which will be subjected to group A inspection shall be preconditioned in accordance with 4.4.

4.3.1 *Group A inspection.*—Group A inspection (including sampling) shall conform to Table II and Standard MIL–STD–105. Group

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A inspection shall be performed in any order which is satisfactory to the Government inspector.

TABLE II *Government group A inspection*

Inspection	Requirement paragraph	Test paragraph	AQL
Visual & Mechanical (Major)		4.5	1%
Visual & Mechanical (Minor)		4.5	4%
Operational Test	3.4	4.6	1% for
Brake Mechanism	3.4	4.7	Group

4.3.2 *Disposition of nonconforming product.*—When defective sample units or rejected lots are resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.4 **Bounce preconditioning.** — After final assembly, each reel unit which will be examined, measured, or tested shall be preconditioned as follows: The reel unit shall be placed in its normal operating position on the table of the Package Tester as made by the L. A. B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The reel unit shall be subjected to this preconditioning for 1 minute.

4.5 **Visual and mechanical inspection.**—The reel unit shall be examined for the defects listed in Table III.

4.6 **Operational test.**—A coil of .109 inch diameter (No. 12 BWG) galvanized iron wire shall be mounted on the reel unit and 500 feet of wire payed out and reeled in again to insure that the unit operates freely without noticeable friction or undue vibration. During this payout of 500 feet of wire, the paying out process shall be stopped quickly to insure that the unit's brake operates satisfactorily and that there is no backlash of wire. The payout speed shall be 10 to 15 miles per hour and at least three stops shall be made to test the automatic brake. Care shall be taken in placing the coil of wire in the reel unit so the wire pays off the coil from the side nearest the brake hinges. For safety, if a vehicle is used for payout, the reel unit shall be securely attached to the vehicle floor.

TABLE III. *Classification of Visual and Mechanical defects.*

Major

1. Upright arms on spool assembly cannot be positioned to meet designated maximum and minimum adjustment.
2. Marking not in accordance with specification.
3. Welds not of proper number or sufficient penetration.
4. Rough edges on metal.

Minor

1. Loose nuts and screws.
2. Missing lockwasher, locknut, or cotter pin.
3. Broken lockwasher, locknut, or cotter pin.
4. Loose rivets.
5. Scratches, cuts, abrasions, etc causing exposure of bare metal.

4.7 **Brake mechanism test.**—The equipment shall be tested to determine that the brake mechanism functions properly when the brake lever arm is operated. This test shall be made for both clockwise and counterclockwise rotation of the reel.

4.8 **Temperature test.** — The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. The equipment shall be tested for compliance with 4.6 and 4.7 at steps 1, 4, 5, 8 and 10 of the cycle.

4.9 **Moisture resistance test.**—The equipment shall be subjected to continuous cycling for five 48 hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. After cycling the equipment shall

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meet the tests of 4.6 and 4.7 and shall be inspected for signs of breakdown in the finish due to moisture penetration. There shall be no evidence of corrosion or impaired paint film

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A—Military package.* — Reel Unit RL–128( )/G, shall be preserved and packaged in accordance with the procedures specified for the designation methods as prescribed in Specification MIL–P–116 and described in the following paragraphs

5.1.1.1 *Reel Unit RL–128( )/G* —Package each reel unit Method III, as follows: Tie and secure all moving and cantilever parts to prevent movement.

5.1.2 *Level B—Limited military package* — Reel Unit RL–128( )/G, shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and described in paragraph 5.1 1

5.1.3 *Level C—Minimum military package* —Reel Unit RL–128( )/G, shall be packaged in accordance with commercial practice and in a manner that will guarantee adequate protection against corrosion deterioration and physical damage during direct shipment to the first receiving activity at lowest transportation rate.

5.2 Packing.

5.2.1 *Level A—Military pack.*—Reel Unit RL–128 ( )/G shall be individually packed in an unsheathed nailed wood crate conforming to the requirements of JAN–P–132. Adequately brace, block, bolt or strap the reel unit within the crate to prevent movement. Fabricate a shroud of waterproof paper conforming to the requirements of JAN–P–125. Place and secure the shroud over the unit so as to withstand high winds and severe storm conditions. Close the crate as specified in the crate specification.

5.2.2 *Level B—Limited military pack.* — Reel Unit RL–128( )/G, shall be individually packed as described in 5.2.1 except the shroud shall not be used.

5.2.3 *Level C—Minimum military pack.*— Reel Unit RL–128( )/G shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification Rules and container specifications for rail shipment, National Motor Freight Classification Rules and container specifications for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation.

5.3 **Marking.**—Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard MIL–STD–129

6. NOTES

6.1 **Intended use.**—Reel Unit RL–128( )/G is used for paying out or reeling in a coil of bare conductor wire during the construction of open wire telephone lines, or a coil of light weight messenger strand for suspending field cable.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Whether domestic or overseas shipment is required.

(c) Samples required.

(d) Marking and shipping of samples.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

MITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE

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